Adiabatic Flame Temperature & Emissions Analysis

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1 Conditions & Data Provenance

To complete this study of the adiabatic flame temperature (T_{ad}) and carbon emissions in the products the fuel chosen was a composition of propane (C_3H_8) and molecular hydrogen (H_2). Fuel blends burning in air at a constant pressure of $p_0 = 1$atm with reactant inlet temperature $T_0 = 298$ K. Molar heat capacities $\bar{c}_p(T)$ and sensible enthalpies were obtained from NASA 7-term polynomial tables and constructed into a matlab structure (`nasa = struct()`).

2 Numerical Model & Equilibrium

Even though the addition of H_2 to the fuel composition made the results more interesting, it was seen the adiabatic flame temperature was closer to T_{ad} for a combustion of pure propane than anticipated. As a consequence, I decided to diversify the blended fuel composition into:

$$100\%C_3H_8; \quad 80\%C_3H_8 + 20\%H_2; \quad 50\%C_3H_8 + 50\%H_2; \quad 20\%C_3H_8 + 80\%H_2; \quad 100\%H_2$$

For $\phi \leq 1.0$, complete oxidation was assumed (CO_2, H_2O, O_2, N_2). For rich conditions ($\phi > 1.0$), equilibrium was closed using the water-gas shift reaction ($CO + H_2O \rightleftharpoons CO_2 + H_2$ and $K_p(T) = \frac{X_{CO_2} X_{H_2}}{X_{CO_2} X_{H_2O}}$) :

$$C_3H_8 + H_2 + a(O_2 + 3.76N_2) \rightarrow bCO_2 + cCO + dH_2O + eH_2 + 3.76aN_2 + fO_2$$

The enthalpy balance $H_{\text{reactants}}(T_0) = H_{\text{products}}(T_{\text{ad}})$ was solved iteratively using Brent's root-finding algorithm (`fzero`) with temperature-dependent NASA 7-coefficient polynomial enthalpies.

3 Results & Key Trends

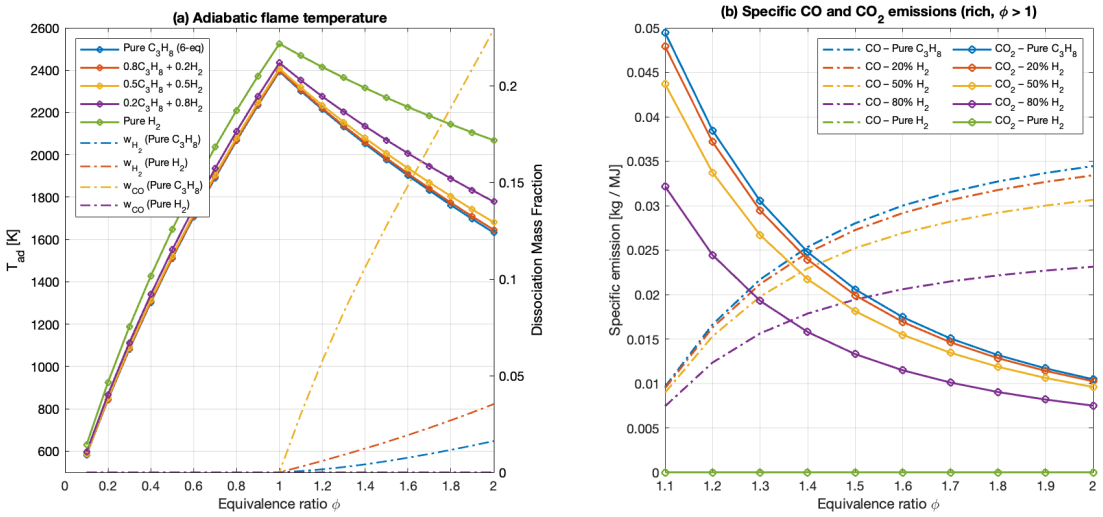


Figure 1: (a) Adiabatic flame temperature T_{ad} and dissociation mass fractions vs. ϕ. (b) Specific CO and CO_2 emissions for rich mixtures ($\phi > 1.0$).

Flame Temperature & Dissociation: T_{ad} peaks near $\phi \approx 1$ (Fig. 1a). Increasing H_2 substitution elevates peak T_{ad} from 2390 K (pure C_3H_8) to 2520 K (pure H_2) due to higher LHV and lower product dissociation penalty. Dissociation mass fractions ($w_{\text{CO}}, w_{\text{H}_2}$) grow rapidly for $\phi > 1$, specially for CO when the fuel composition was 100% propane.

Specific Emissions ($\phi > 1.0$): As ϕ increases from 1.1 to 2.0 (Fig. 1b), incomplete combustion drops CO_2 emissions (0.05 → 0.01 kg/MJ) while CO surges (0.01 → 0.035 kg/MJ for pure propane). H_2 addition significantly reduces specific CO emissions, reaching zero for pure H_2. This is important as it proves that an addition of H_2 to the fuel blend reduces pollutant components in the products. Evidently, this is a simple study but it raises the question of whether this procedure will be possible in the future or not, once the variables that are now alarming reflect a harmless attitude.

References

[1] B. McBride et al., NASA RP-1311: *Coefficients for Calculating Thermodynamic Properties*, 1993.